Abdullah Al-Nafisah

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Education

MSc, Electrical and Computer Engineering 2022 – 2024
King Abdullah University of Science and Technology (KAUST), Thuwal, KSA | GPA 3.83/4.0 | Thesis: *Turbo-Equalized FTN-Inspired Modulations*, doi:10.25781/KAUST-M5140

BSc, Double Major in Electrical Engineering & Physics 2013 – 2019
King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, KSA | GPA 3.76/4.0, 1st Honors

Exchange Student Fall 2016
Georgia Institute of Technology, Atlanta, GA, USA | GPA 3.5/4.0

Summer School on Superconducting Electronics Summer 2024
FLUXONICS, The European Foundry for Superconducting Electronics e.V., Corsica, France

Technical Skills

Languages Python, C++, C, Verilog/SystemVerilog
Verification cocotb, RTL simulation and regression
Tooling Tcl (Vivado, Quartus, Synopsys, Cadence), CMake/Make, Git

Research Experience

Research & Development Oct 2020 – Present
National Company of Telecommunications and Information Security (NTIS), KSA
– Developed direct RF sampling architectures with x16 DACs and x16 ADCs using an AMD/Xilinx RFSoC board.
– Developed and worked on different optical communication systems using an AMD/Xilinx MPSoC board.
– Developed PCB designs and FPGA-based (VHDL and Verilog) implementations for controllers and custom UART, I2C, and SPI drivers for embedded chips.

Master's Thesis Research Spring 2023 – Spring 2024
Supervised by Dr. T. Al-Naffouri and Dr. M. Siala at KAUST, Thuwal, KSA
– Developed optimized Python scripts for efficient decoding algorithms to obtain BER plots and EXIT charts.
– Integrated FTN-inspired modulations with turbo equalization and LDPC decoding to improve the BER performance and achieve superior results to those reported in state-of-the-art literature.

Guided Research on Analog Electronic Circuits Fall 2018 – Spring 2019
Supervised by Dr. M. Abuelma'atti at KFUPM, Dhahran, KSA
– Developed a memristor-based chaotic masking system built from off-the-shelf components and experimentally validated the synchronization of two chaotic oscillators.
– Developed a novel design of a tunable comb filter (based on CFOAs).

Mitacs Globalink Research Internship Summer 2018
Supervised by Dr. S. Valluri at Western University, London ON, Canada
– Investigated the Lambert W function to describe the I-V characteristics of diode-based circuit models.

Visiting Student Research Program (VSRP) Internship Summer 2017
Supervised by Dr. H. Fariborzi at KAUST, Thuwal, KSA
– Applied the Landau-Lifshitz-Gilbert (LLG) equations in Cadence using Verilog-A for All Spin Logic (ASL) circuits.
– Developed compact modules for charge/spin conduction, including ferromagnets (FM), non-magnets (NM), and FM/NM interfaces, along with generalizing the rotational matrix and optimizing the LLG solver.

Publications & Posters

Papers

1. M. Siala, A. Al-Nafisah, and T. Al-Naffouri, "Nyquist Signaling Modulation (NSM): An FTN-Inspired Paradigm Shift in Modulation Design for 6G and Beyond," *arXiv:2511.08553*, 2025.
2. A. Al-Nafisah, D. Khan, S. Amara, and Y. Massoud, "Synchronization of Two Transistor-Based Chaotic Oscillators: Experimental Verification," *IEEE 65th International Midwest Symposium on Circuits and Systems (MWSCAS)*, Fukuoka, Japan, pp. 1–4, 2022.
3. M. T. Abuelma'atti and A. Al-Nafisah, "New CFOA-Based Shadow Analog Comb Filter," *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 1–8, 2020.
4. M. T. Abuelma'atti and A. Al-Nafisah, "Synchronization of Memristor-Based Chaotic Oscillator: Experimental Verification," *Journal of Active and Passive Electronic Devices (JAPED)*, vol. 15, no. 1–2, pp. 21–28, 2020.
5. M. T. Abuelma'atti and A. Al-Nafisah, "A Memristor-Based Chaotic-Masking for Analog Spread-Spectrum Communication," *Indonesian Journal of Electrical Engineering and Computer Science (IJEECS)*, vol. 14, no. 2, pp. 966–971, 2019.

Posters

1. A. Al-Nafisah, M. Siala, and T. Al-Naffouri, "Utilizing EXIT Charts for Turbo Processing of M-ASK Modulations in 6G Communications," *6G Summit*, Abu Dhabi, UAE, 2023.
2. A. Al-Nafisah, F. Al-Masri, F. Swaid, M. Al-Mousa, and M. Al-Shams, "Prototype of a Wireless Seismic Data Acquisition System," *Senior Design Project, KFUPM*, Dhahran, KSA, 2019.
3. A. Al-Nafisah, N. Iqbal, and S. Al-Dharrab, "Throughput Analysis of Wireless Geophone Networks for Seismic Surveys," *IEEE GCC Conference & Exhibition*, Bahrain, 2017.
4. A. Al-Nafisah, M. Alawein, and H. Fariborzi, "Modeling and Simulation of All-Spin Logic Device," *KAUST Visiting Student Research Program (VSRP)*, Thuwal, KSA, 2017.